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Práctica #1: Descripción y simulación con Cadence

En esta práctica se debe repetir el proceso mostrado en el tutorial, pero cambiando la compuerta AND por cualquier otra compuerta de 2 entradas.

Código de descripción de hardware

```
1 //Verilog HDL for "Verilog", "AND_2x" "functional"
2
3 module AND_2x (A,B,Z);
4     input  A,B;
5     output [6:0]Z;
6
7     assign Z[0]= ~A;
8     assign Z[1]= A & B;
9     assign Z[2]= ~(A & B);
10    assign Z[3]= A ^ B;
11    assign Z[4]= A ~^ B;
12    assign Z[5]= A | B;
13    assign Z[6]= ~(A | B);
14
15 endmodule
16
```

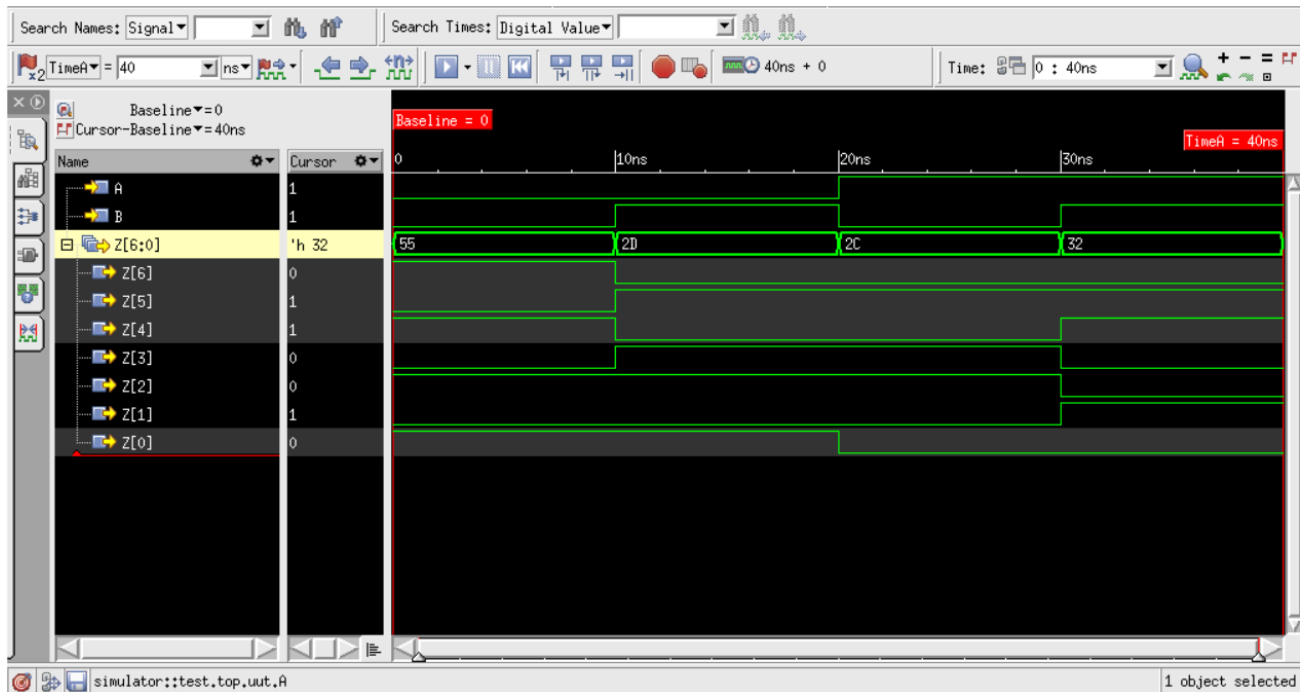
Código de test bench

```
1 //Verilog HDL for "Verilog", "AND_2x_TB" "functional"
2
3 module AND_2x_TB ( );
4
5     reg  A,B;
6     wire Z;
7
8     AND_2x uut (A,B,Z);
9
10    initial begin
11
12        A = 1'b0;
13        B = 1'b0;
14        #10
15
16        A = 1'b0;
17        B = 1'b1;
18        #10
19
20        A = 1'b1;
21        B = 1'b0;
22        #10
23
24        A = 1'b1;
25        B = 1'b1;
26        #10
27
28        $display("Test completed");
29
30    end
31
32 endmodule
```

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Simulación



CODIGOS

//Verilog HDL for "Verilog", "AND_2x" "functional"

```
module AND_2x (A,B,Z);
    input A,B;
    output [6:0]Z;

    assign Z[0]= ~A;
    assign Z[1]= A & B;
    assign Z[2]= ~(A & B);
    assign Z[3]= A ^ B;
    assign Z[4]= A ^^ B;
    assign Z[5]= A | B;
    assign Z[6]= ~(A | B);
```

endmodule

////////////////////////////////////

//Verilog HDL for "Verilog", "AND_2x_TB" "functional"

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```
module AND_2x_TB ( );

    reg A,B;
    wire Z;

    AND_2x uut (A,B,Z);

    initial begin

        A = 1'b0;
        B = 1'b0;
        #10

        A = 1'b0;
        B = 1'b1;
        #10

        A = 1'b1;
        B = 1'b0;
        #10

        A = 1'b1;
        B = 1'b1;
        #10

        $display("Test completed");

    end

endmodule
```